

EX10:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.ensemble import AdaBoostClassifier
df = pd.read_csv(r"C:/Users/24ucs152/Downloads/diabetes_data.csv")
print("----- DATASET -----")
print(df.head(10).to_string(index=False))
X = df[["Age", "BMI", "Blood Pressure"]]
y = df["Diabetes"]
model = AdaBoostClassifier(n_estimators=50, random_state=42)
model.fit(X, y)
age = float(input("\nEnter Age: "))
bmi = float(input("Enter BMI: "))
bp = float(input("Enter Blood Pressure: "))

new_patient = pd.DataFrame(
    [[age, bmi, bp]],
    columns=["Age", "BMI", "Blood Pressure"]
)
prediction = model.predict(new_patient)
print("\nPredicted Class:", prediction[0])
if prediction[0] == 1:
    print("Patient is at RISK of Diabetes")
else:
    print("Patient is NOT at RISK of Diabetes")
```

output:

```
----- DATASET -----
Age  BMI  Blood Pressure  Diabetes
25  22.0      110         0
30  24.0      115         0
35  27.0      120         0
45  31.0      135         1
50  34.0      140         1
60  20.2      148         1
35  22.5      137         0
54  19.7      141         1
21  19.9      123         0
55  22.0      146         1

Enter Age: 42
Enter BMI: 30
Enter Blood Pressure: 130

Predicted Class: 0
Patient is NOT at RISK of Diabetes
> |
```

Age	BMI	Blood Pressure	Diabetes
25	22.0	110	0
30	24.0	115	0
35	27.0	120	0
45	31.0	135	1
50	34.0	140	1
60	20.2	148	1
35	22.5	137	0
54	19.7	141	1
21	19.9	123	0
55	22.0	146	1
54	26.4	147	1
37	34.2	140	0
64	26.5	150	1
33	37.1	144	1
48	22.1	139	0
31	34.8	142	0
52	36.9	152	1
23	37.3	139	0
27	33.0	137	0
61	28.6	152	1
33	26.7	133	0
40	24.3	136	0
21	28.7	127	0
58	28.8	150	1
43	26.8	138	0
71	29.8	162	1
37	21.8	132	0
38	20.8	133	0
55	30.3	150	1
29	26.5	129	0
28	34.9	136	0
60	21.9	146	1
22	23.3	123	0
42	32.3	144	1
58	35.1	154	1
65	37.7	160	1
67	31.7	159	1
73	20.0	163	1